SEAGAS - Team Update: What Is Done, What Is Left, What To Do Next

From Carl (Soh Way Miin), Backend Developer, to Group 10 | 1 November 2026
COS40005 Computing Technology Project A | POC004

In short: today we finished the last items from the lecturer's feedback of 22 October. **Advisors** now manage the **skills library**, job roles and learning resources of their own courses, and the **skill-gap report** can be filtered by **programme** and by **skill category**, with a soft-skills view. Scores now say **"Not required"** or **"No data yet"** instead of a misleading 0%. **My Skills** is grouped by category with one colour per category, Progress Tracking has a **Run again** button for reassessment, and the **Job Readiness Score** formula is written up and justified with a sensitivity analysis. All the website functions planned for Sprint 2 are now built. After you pull, run **start_all.bat** once (section 2.1).

Note: these changes went straight to dev (commit `cd66917`), without a pull request. They pass the automated tests (185 backend tests) but have **not been checked in a browser yet**; see section 2.2.

1. What Is Done (1 November)

1.1 Advisors

Feature	What it does
Skills library for advisors	Advisors now have **Skills & Roles** and **Learning Resources** in their menu (the same pages as the admin's). The lecturer asked to move skills-library management from the admin to the advisor. • Advisors add skills and job roles **for their own courses** only. • They can edit or switch off an entry when **all of its courses are theirs**. • They add learning resources for any skill their students see, and can change the resources they added. • Entries shared with other courses (for example the soft skills used by every course) stay with admins and show "Shared · admin only", so a Business Head of Programme cannot switch off a skill that Computer Science students use. The API moved from `/api/admin/skills|roles|resources` to `/api/library/...` (`backend/routers/library.py`).
Skill gaps by programme and category	Advisor Dashboard → Skill Gaps has a **programme (major)** filter next to the job-role filter. Four cards (Technical, AI & Digital, Analytical, **Soft skills**) show the Strong / Developing / Gap share of each category for the selected students. Clicking a card lists only that category's skills. This answers the lecturer's request for a soft-skill focus by programme.

1.2 Students and advisors: scores and skills

Feature	What it does
"Not required" and "No data yet"	The seven dimension bars (student Dashboard, Skill Gap page, advisor's student page) now show three cases: • **Not required**: the job asks for no skill of this category. • **No data yet**: nothing is recorded in the profile (no projects, certifications or work experience, or no skill of that category listed). A short note says what to add. • **0%**: only a real result, when the student does list skills of that category but none fits the job. The lecturer asked how a zero should be read when it may mean missing information.
My Skills grouped by category	My Profile → Skills shows four groups, **one colour per category**. Levels are shown as dots (●○○ beginner, ●●○ intermediate, ●●● advanced) instead of red / amber / green. The same colours are used on Export CV and on the advisor's student page.
Run again (reassessment)	Progress Tracking has ↻ **Run again**. It assesses the current profile against the same role or pasted job description, with the same method, and shows the change. When the student has completed activities since the last assessment, a reminder asks them to add the new skills and run it again (S2-13-03).

1.3 Job Readiness Score write-up

The lecturer asked us to research and justify the readiness formula. `docs/JOB_READINESS_SCORE.md` covers:

- the formula with a worked example;
- why the importance weights are 3 / 2 / 1 and why Developing gets half credit;
- why the readiness bands are 70 / 40;
- the seven dimensions and why the experience dimensions are not part of the score (to avoid counting the same evidence twice);
- the zero / "No data yet" rule;
- a first mapping to the MQF 2.0 clusters, and the limitations.

`data/jrs_sensitivity.py` re-scores 5,001 assessments with other weights. The student ranking hardly changes (Spearman ρ ≥ 0.987), so comparisons do not depend on the exact weights:

Scheme	Mean	Spearman ρ vs current	Students changing band
Current 3/2/1, developing 0.5	48.3	1.000	0 (0.0%)
Equal 1/1/1	42.9	0.987	953 (19.1%)
Steeper 4/2/1	50.4	0.998	325 (6.5%)
Developing 0.25	44.8	0.994	479 (9.6%)
Developing 0.75	51.9	0.993	491 (9.8%)
Required skills only	61.6	0.949	1,959 (39.2%)

These figures use the assessments made before the matching-rule update of 30 October. They will be measured again after `start_all.bat` has re-assessed the students.

1.4 Also since the last update (Yong Hin, 30 October)

Change	What it does
Skill evidence from projects and certifications (PR #21)	At assessment time, the student's projects (technologies, title, description) and certifications count as evidence for the matching skills. A beginner skill used in a listed project can now be Strong; the match reason names the project. On the synthetic students, mean readiness rose by 1.02 and 1.7% of skill rows moved from Developing to Strong. Matching-rule version `2026-10-30`, so `start_all.bat` re-assesses everyone.
350 job descriptions (PR #22)	The JD dataset grew to 350 (35 per role, 10 roles). 285 come from the Kaggle LinkedIn Job Postings dataset (CC BY-SA 4.0), cleaned and anonymised. Business role rules were added, and a new region column was added. The 60 gold-set JD references are unchanged.

1.5 Database and tests

- New column `recommendations_library.created_by` (who added a resource), in `schema.sql` and `migrate_existing_db.sql`.
- Backend tests: **185 passed**, including new tests for the advisors' library scope, the skill-gap filters and the category summary, and Run again.
- NLP tests: 355 passed, **1 failed**: `test_sample_jds_smoke[kaggle/3]`. That sample is a security-guard job added on purpose to test the title filter; it has no IT skills, so the smoke test's "every sample finds a skill" check fails. The test needs fixing, not the website (Yong Hin, section 4).

2. What To Do Next

2.1 After you pull (every member)

1. `git pull` on dev.
2. Run **start_all.bat** once. It:
   - adds the new database column (without it the Learning Resources page fails);
   - loads the 350 job descriptions;
   - re-assesses the 5,000 synthetic students with the 30 October rules (this takes a few minutes).
3. Carl then runs `python data/jrs_sensitivity.py` and updates sections 2.4 and 2.5 of `docs/JOB_READINESS_SCORE.md` with the new figures.

2.2 Please check in the browser (Ren Hang, Shanice)

- **Advisor → Skills & Roles / Learning Resources**:
  - sign in as a Head of Programme;
  - add a skill, a role and a resource;
  - check that shared entries show "Shared · admin only" and that the course boxes only offer your courses.
- **Advisor Dashboard → Skill Gaps**: programme filter, the four category cards, clicking a card, CSV export.
- **Student Dashboard and Skill Gap page**: "Not required" / "No data yet" for a student with no projects or no soft skills.
- **My Profile → Skills**: grouped colours, removing a skill, CV upload.
- **Progress Tracking**: Run again, and the reminder after completing an activity.
- Narrow window widths: no text cut off, tables scroll inside their card.

3. Sending Real Emails (one-time setup, not done yet)

Without a mail server, welcome emails and password emails are only stored in Student Intake → Email outbox. For the demo they should really arrive. **Who:** Carl, on the computer that runs the backend for the demo.

1. Prepare a Gmail account for the demo and turn on **2-Step Verification**.
2. Google Account → Security → **App passwords**: create a 16-letter App Password (the normal password does not work).
3. Add these lines to `backend/.env` (this file is **not** in git — never commit it):

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_SECURITY=starttls
SMTP_USER=your.mailbox@gmail.com
SMTP_PASSWORD=the 16-letter App Password
SMTP_FROM=your.mailbox@gmail.com
```

4. Restart the backend.
5. Sign in as an admin, open **System Settings**, check that it says a mail server is configured, and press **Send test email** (look in the spam folder too).
6. Go to **Student Intake** → **Email outbox** and **Resend** the emails that were stored before.

The full steps are also in `docs/SEAGAS_TeamUpdate_30October_By_Waymiin.pdf`. Do not share the App Password in chat or in the report; anyone with it can send mail as the account.

4. What Is Still Left

The website functions planned for Sprint 2 are all built. What remains is setup, testing, research and the report.

Owner	Task	When
Carl	Mail-server setup (section 3); run start_all.bat and update the sensitivity figures (section 2.1).	Before demo
Yong Hin	Fix the NLP test `test_sample_jds_smoke[kaggle/3]` by leaving the non-IT sample out of that test. Check the Business role templates.	This week
Aaron	Check the Business learning resources.	This week
Ren Hang, Shanice	Browser check of all new pages (section 2.2; S2-05, S2-10, S2-10-01, S2-10-02).	This week
Ren Hang	Admin interface UI/UX refinement (S2-09-01).	Sprint 2
Shanice	Student Dashboard information hierarchy (S2-14-01).	Sprint 2
Aaron, Shanice	Update the Sprint 2 report. Sections 2.0.10, 2.0.12 and 3.0.4, Table 1 and Table 3 still list items as "Not Started" that are now done: • S2-06 CV upload; • S2-13-01 progress states and S2-13-03 reassessment; • S2-14-02 real data on the Student Dashboard; • S2-15, S2-15-01 and S2-15-02 advisor cohort views; • the feedback items from 22 October. `docs/JOB_READINESS_SCORE.md` can be used for the readiness-score justification.	Before submission

Whole team	Research: • check MQF 2.0 skill by skill and look at Malaysian job competency surveys (section 4 of the JRS document is only a first mapping). Evaluation: • test with 25 UAD students; • run the SUS usability test. Also prepare the demo and keep following up on an industry partner.	Sprint 3-4
Optional (Carl)	Make the score colours the same everywhere. Student pages colour readiness at 70 / 50, while the advisor pages use the System Settings bands (70 / 40 by default). A few minutes' work.	Any time